

Computer Networks
Lecture 12

Shmulik Kaplan

Today

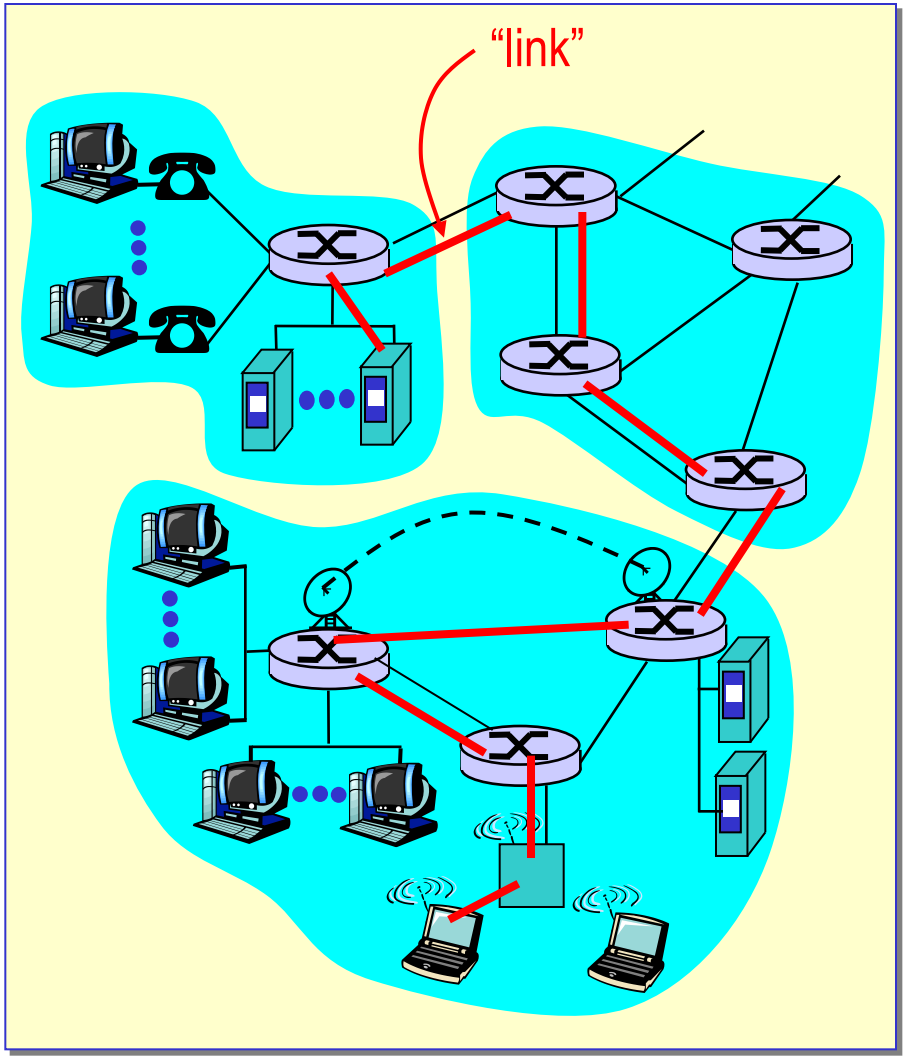
- Link Layer – Introduction
- Error detection, error correction
- Multiple access links
- LAN
  - Addresses
  - Routing
  - ARP
- Routing between to a different LAN
- Ethernet
- Privacy note

Outline

Data-link layer has responsibility of transferring packets from one node to adjacent node over a link

Topics

- Error detection and correction
- Multiple Access Protocols
- LAN addresses (MAC address)
- ARP:Address Resolution Protocol
- Ethernet
- Hubs, bridges, and switches



Link layer

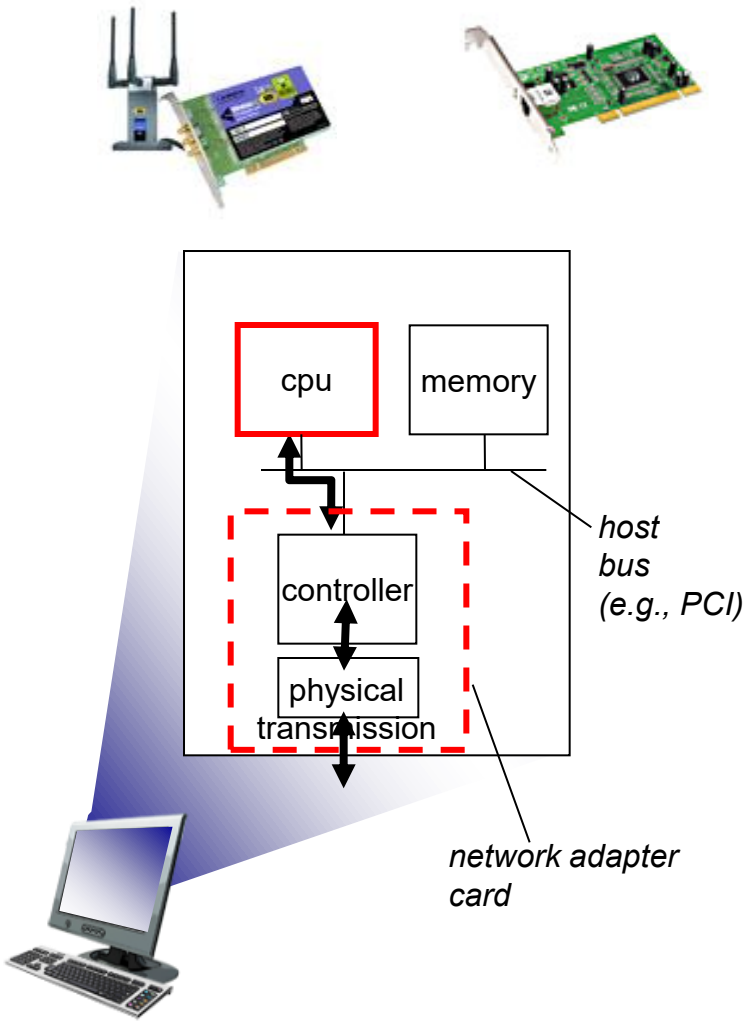
- Datagram transferred by different link protocols over different links:
  - e.g., Ethernet, WiFi.
- Each link protocol provides different services

Transportation analogy

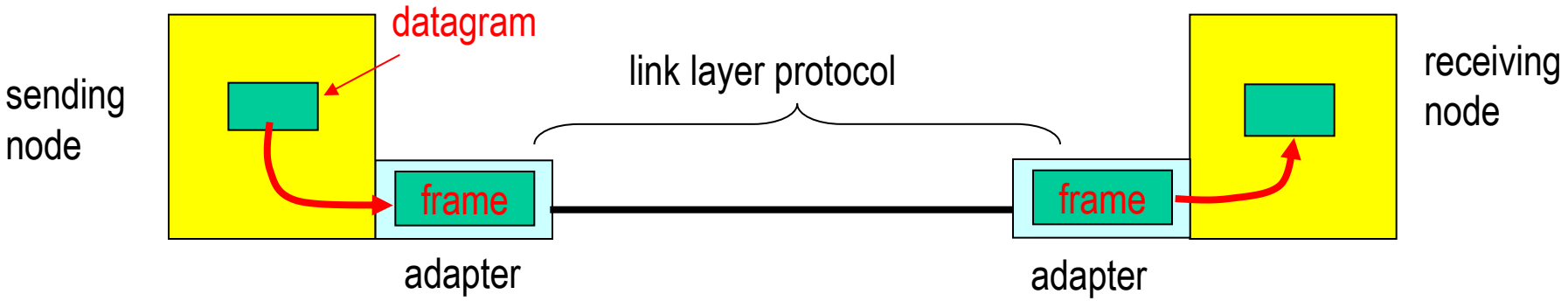
- trip from Princeton to Lausanne
  - limo: Princeton to JFK
  - plane: JFK to Geneva
  - train: Geneva to Lausanne
- tourist = datagram
- transport segment = communication link
- transportation mode = link layer protocol
- travel agent = routing algorithm



Where is the link layer implemented?



Adaptors Communicating







Link Layer Services

- Framing, link access
  - encapsulate datagram into frame, adding header, trailer
  - channel access if shared medium
- Error Detection & filtering
  - receiver detects presence of errors: signals sender for retransmission or drops frame
- Error Correction
  - receiver identifies *and corrects* bit error(s) without resorting to retransmission

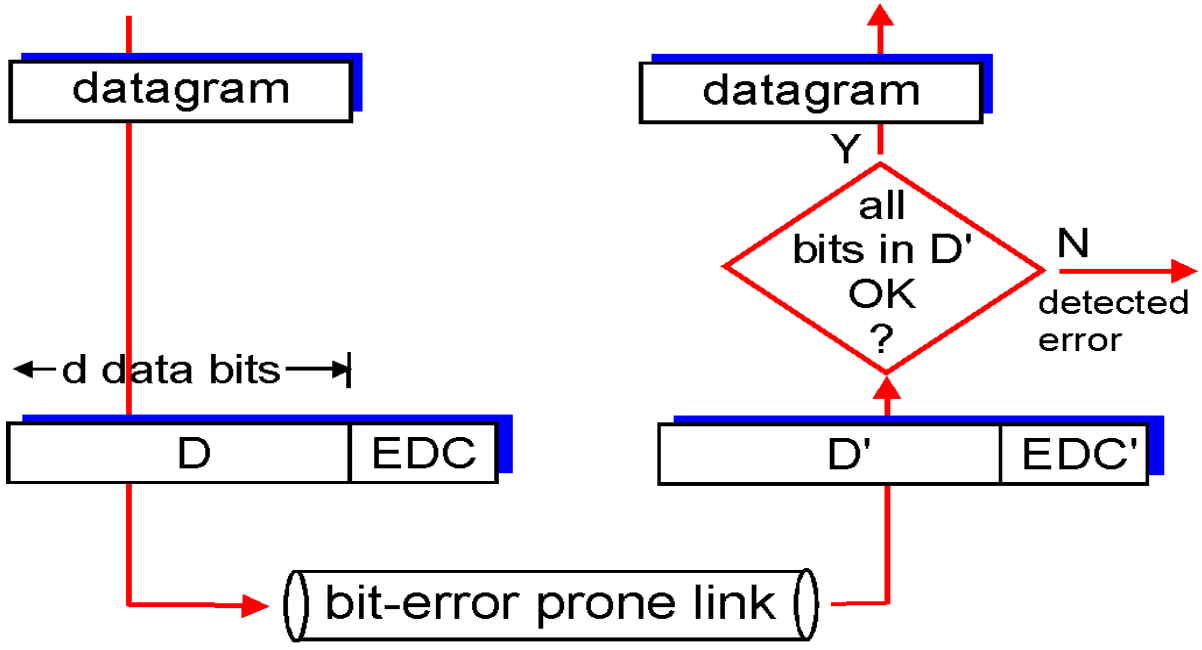
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Error Detection

EDC= Error Detection and Correction bits (redundancy)
D = Data protected by error checking, may include header fields

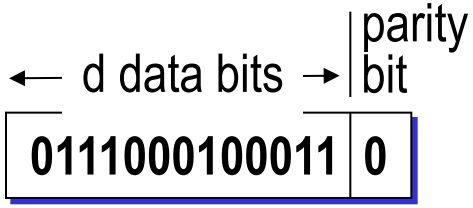
Error detection not 100% reliable!



Parity Checking

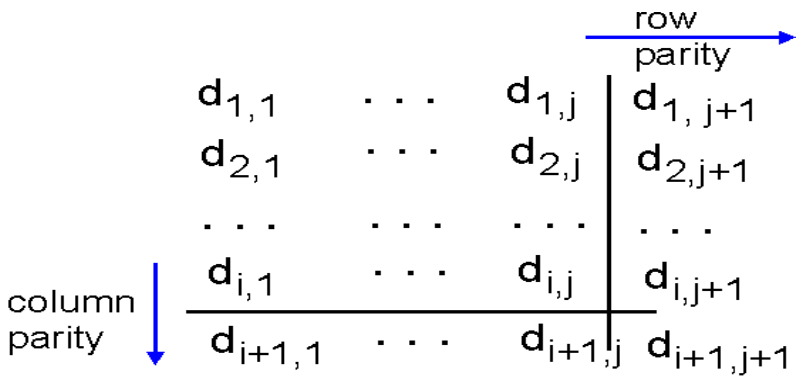
Single Bit Parity:

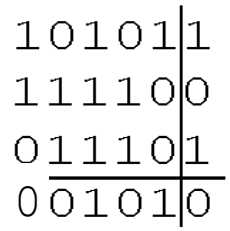
Detect single bit errors



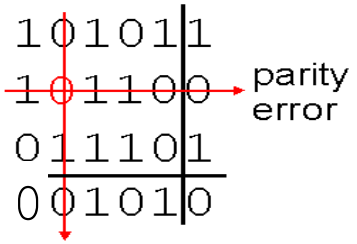
Two Dimensional Bit Parity:

Detect *and correct* single bit errors





no errors



correctable single bit error

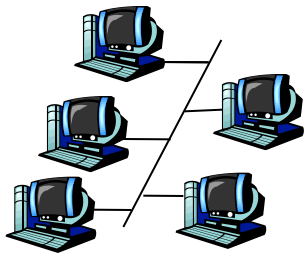
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Multiple Access Links and Protocols

Two types of links

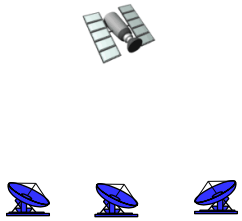
- point-to-point
  - PPP for dial-up access
  - point-to-point link between Ethernet switch and host
- broadcast (shared wire or medium)
  - traditional Ethernet
  - 802.11 wireless LAN (Wi-Fi)



shared wire (e.g., cabled Ethernet)



shared wireless (e.g., 802.11 WiFi)



satellite



humans at a cocktail party (shared air, acoustical)

Multiple Access Protocols

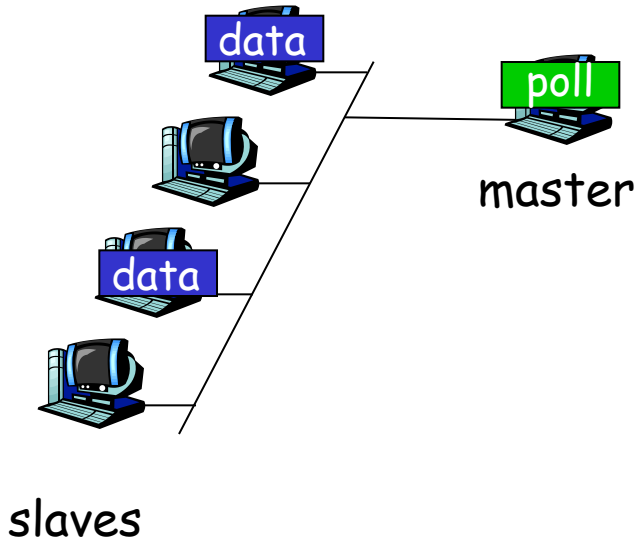
Three broad classes

- Channel Partitioning
  - TDMA: Time Division Multiple Access
  - FDMA: Frequency Division Multiple Access
  - CDMA: Code Division Multiple Access,
- Random Access: "recover" from collisions
  - ALOHA, Slotted-ALOHA
  - CSMA: Carrier Sensing Multiple Access
  - **CSMA/CD used in Ethernet** (Collision Detection)
- "Taking turns"
  - polling from a central site,
  - token passing

“Taking Turns” MAC protocols

Polling:

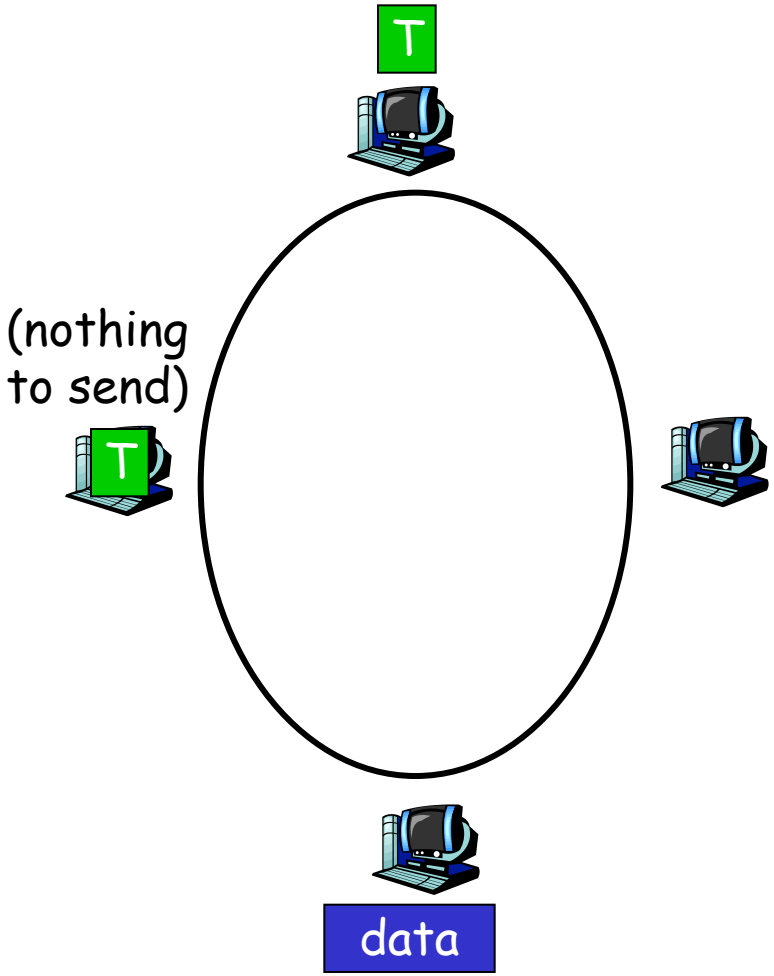
- master node “invites” slave nodes to transmit in turn
- typically used with “dumb” slave devices
- concerns:
  - polling overhead
  - latency
  - single point of failure (master)



“Taking Turns” MAC protocols

Token passing:

- control **token** passed from one node to next sequentially.
- token message
- concerns:
  - token overhead
  - latency
  - single point of failure (token)

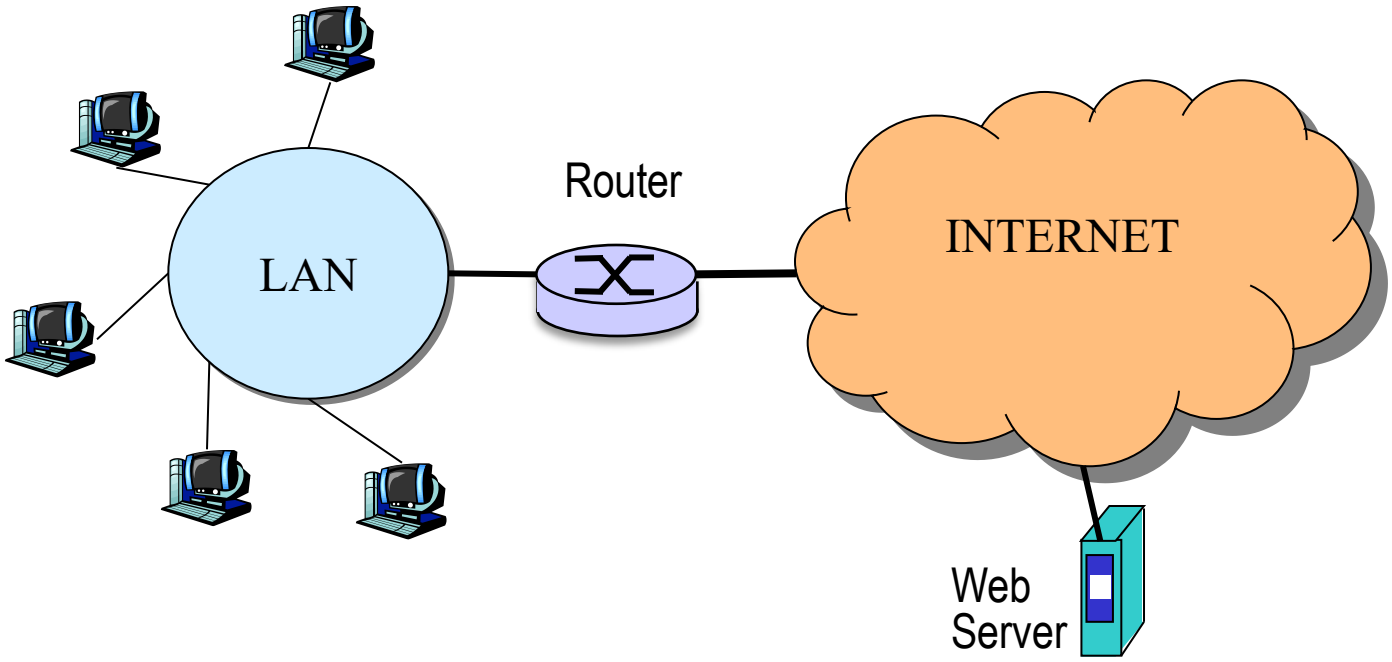


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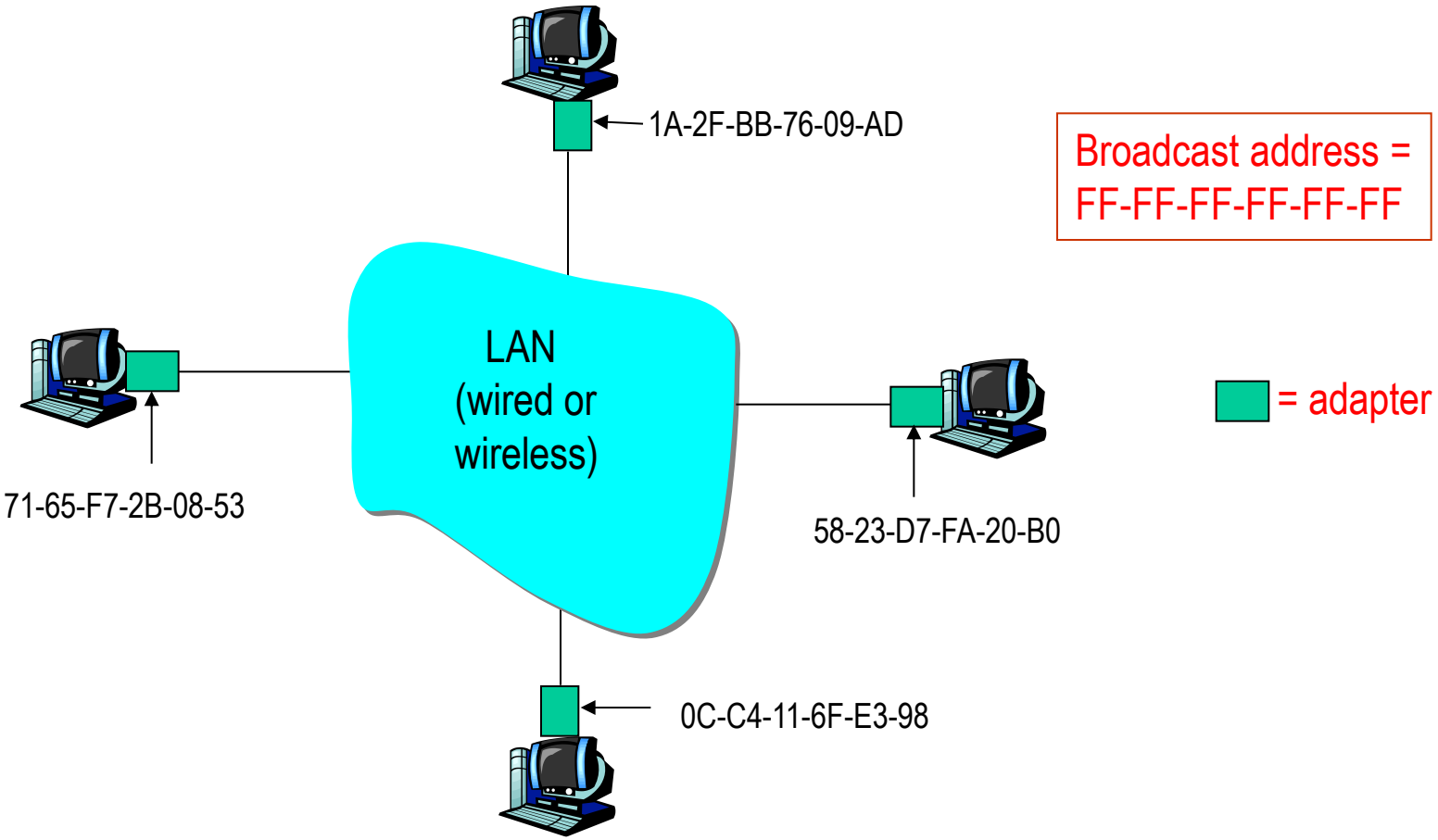
LAN technologies

- addressing
- Ethernet
- hubs, bridges, switches

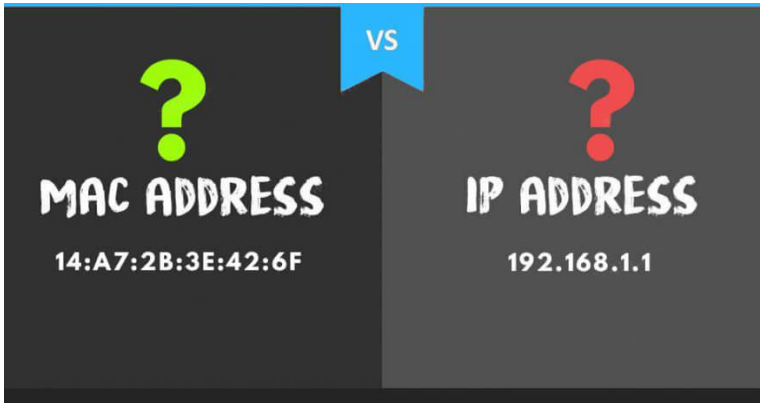


LAN Addresses

Each adapter on LAN has unique LAN address



LAN Addresses

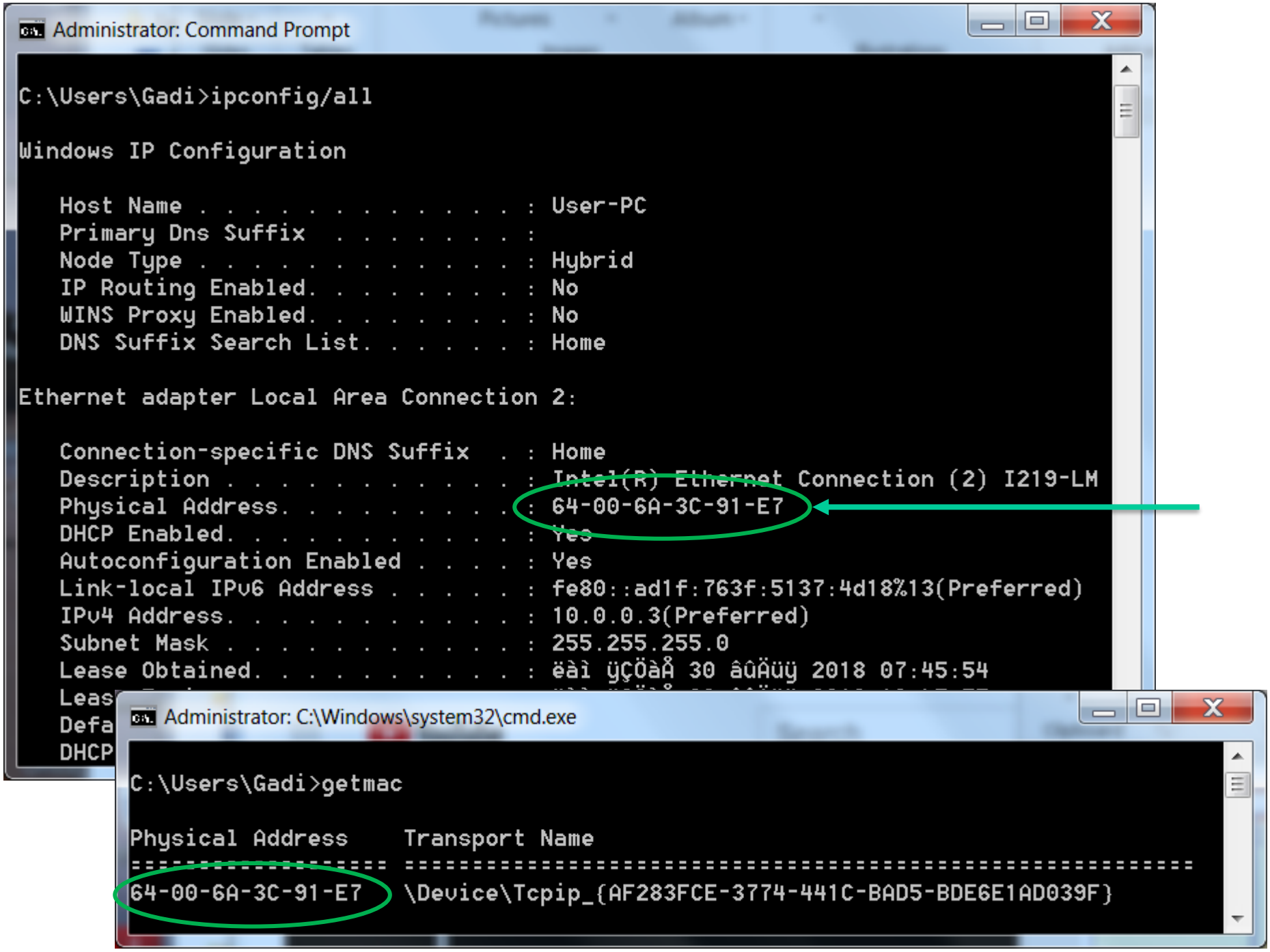


LAN address

- Used to get datagram from one interface to another physically-connected interface (same network)
- 48 bit address (for most LANs) burned in the adapter ROM
- Also called
  - physical address
  - Ethernet address
  - **MAC** (Media Access Control) address

32-bit IP address:

- *network-layer* address
- used to get datagram to destination IP network (recall IP network definition)


Administrator: Command Prompt
C:\Users\Gadi>ipconfig/all
Windows IP Configuration
Host Name . . . . . : User-PC
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Home
Ethernet adapter Local Area Connection 2:
Connection-specific DNS Suffix . : Home
Description . . . . . : Intel(R) Ethernet Connection (2) I219-LM
Physical Address. . . . . : 64-00-6A-3C-91-E7
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::ad1f:763f:5137:4d18%13(Preferred)
IPv4 Address. . . . . : 10.0.0.3(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : ëàì ÿÇÖàÅ 30 âûÄüÿ 2018 07:45:54
Administrator: C:\Windows\system32\cmd.exe
C:\Users\Gadi>getmac
Physical Address Transport Name
64-00-6A-3C-91-E7 \Device\Tcpip_{AF283FCE-3774-441C-BAD5-BDE6E1AD039F}

LAN Addresses

- MAC address allocation administered by IEEE
  - manufacturer buys portion of MAC address space
- Analogy:

  (a) MAC address: like ID Number

  (b) IP address: like postal address
- MAC flat address => portability
  - can move LAN card from one LAN to another
- IP hierarchical address NOT portable
  - depends on IP network to which node is attached

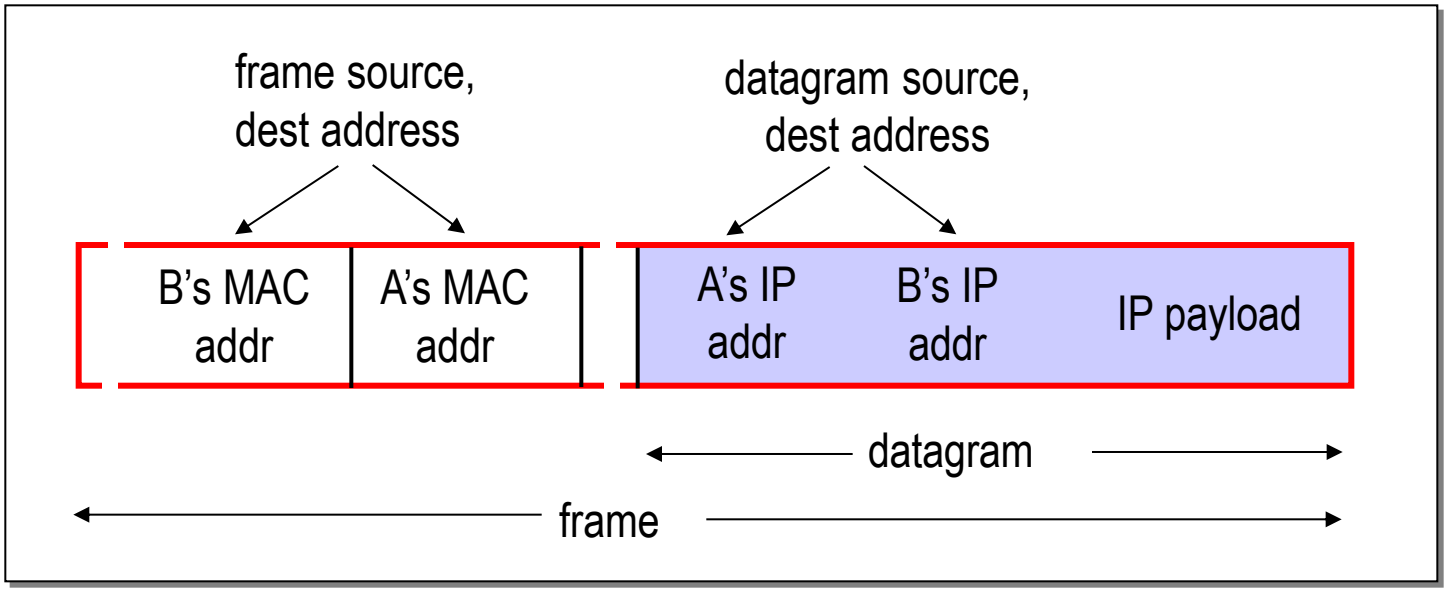
Why both IP and MAC addresses?
Why don't eliminate the MAC and use just IP ?

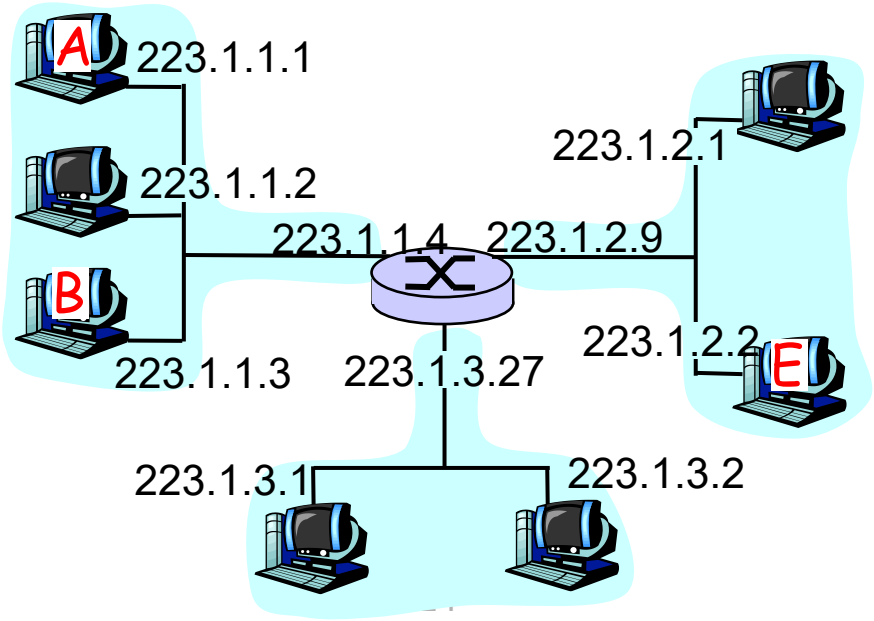
- There isn't anything stopping you from designing network which has just IP address for both L2 and L3 …
- However this network would only ever support IP
  - You also couldn't run various other protocols or some new protocol you're developing..
  - when next-generation of IPvX comes, it would not work
- Now you could have ran same Ethernet switches and migrated your network from IPv4 to IPv6, without touching your LAN at all.
- Abstraction layers increase development speed and encourage innovation.

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Recall earlier routing discussion



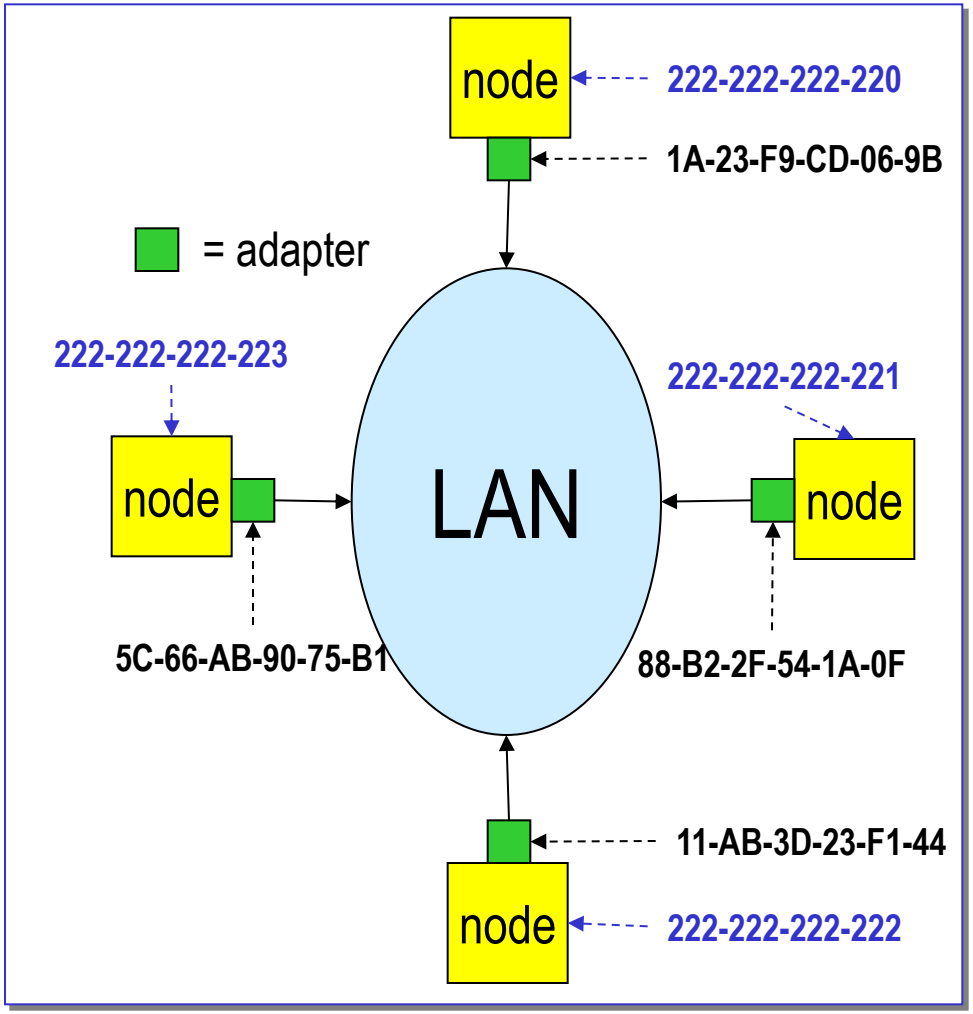


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ARP: Address Resolution Protocol

Question: how to determine MAC address of B knowing B's IP address?



- Each IP node (Host, Router) on LAN has ARP table
- ARP Table: IP/MAC address mappings for some LAN nodes

  < IP address; MAC address; TTL>

  - TTL (Time To Live): time after which address mapping will be forgotten (typically 20 min)

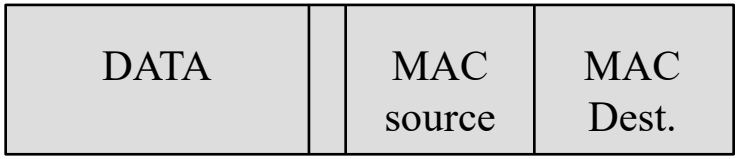
IP	MAC	TTL

ARP



- A wants to send datagram to B, and A knows B's IP address.
- Suppose B's MAC address is not in A's ARP table.
- A broadcasts ARP query packet, containing B's IP address
  - all machines on LAN receive ARP query
- B receives ARP packet, replies to A with its (B's) MAC address
  - frame sent directly to A's MAC address

- A caches (saves) IP-to-MAC address pair in its ARP table until information becomes old (times out)
- ARP is "plug-and-play":
  - nodes create their ARP tables without intervention from net administrator



To see the ARP table on your computer run "ARP –a" from the command prompt

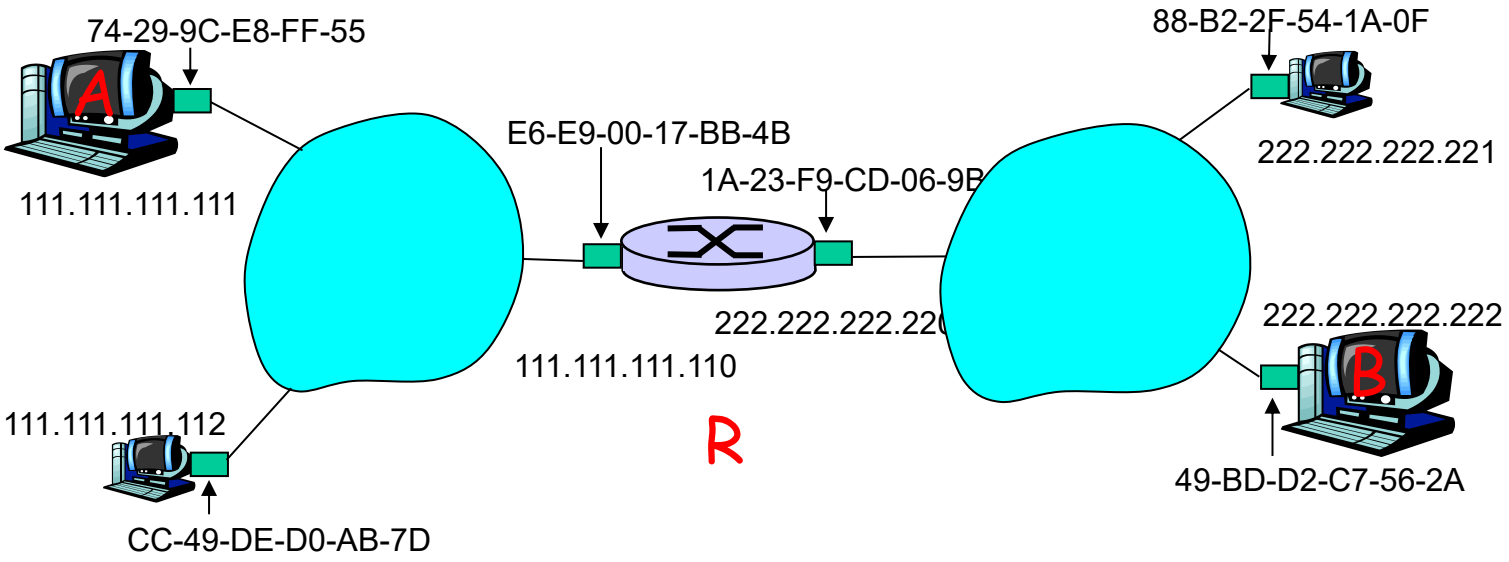
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Addressing: routing to another LAN

Walkthrough: send datagram from A to B via R

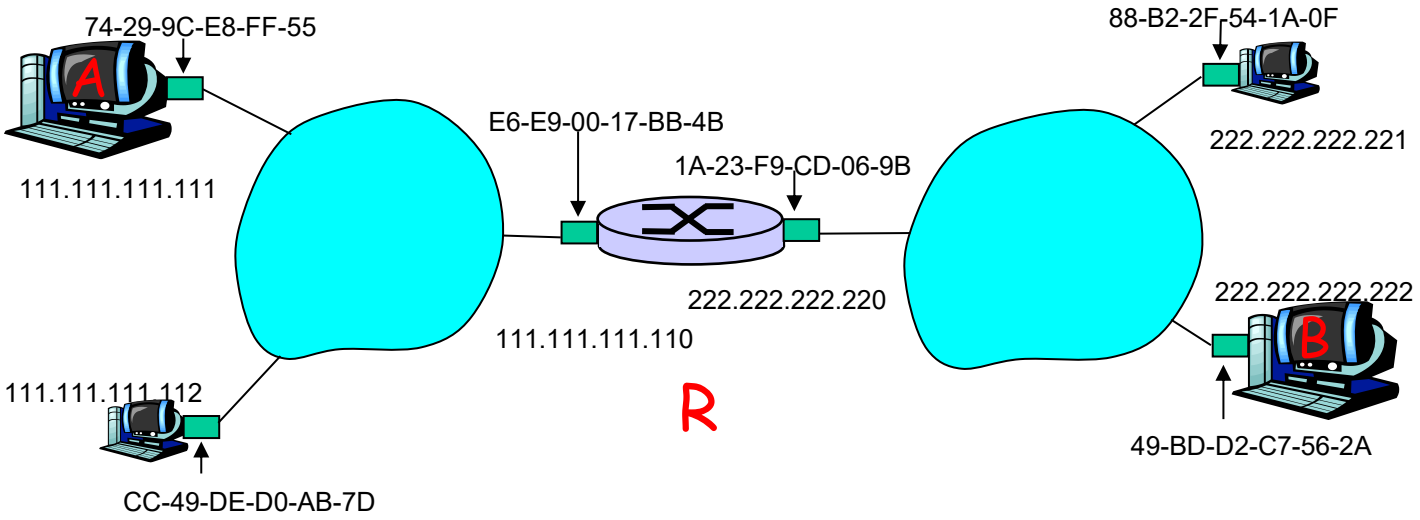
assume A knows B's IP address



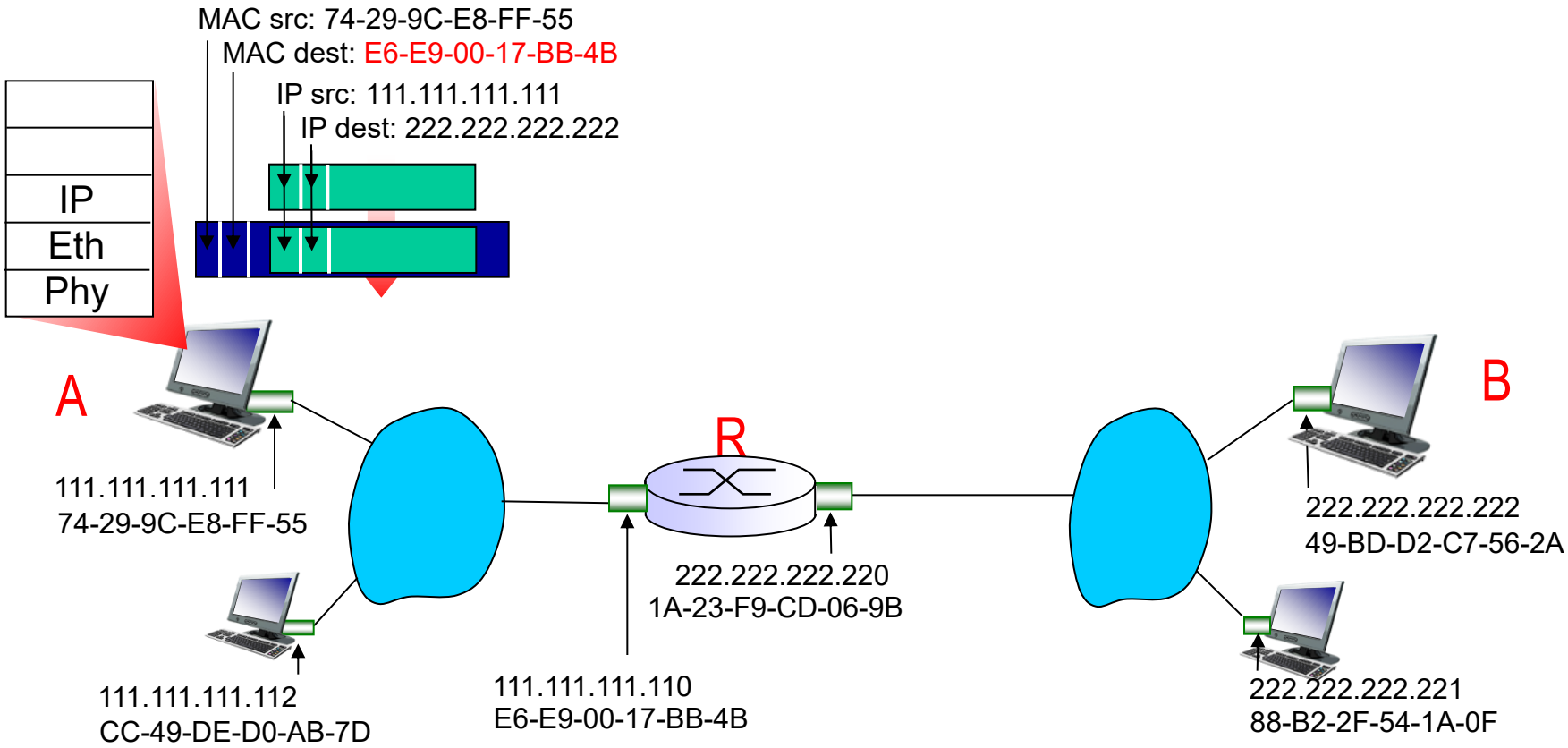
- **Two** ARP tables in router R, one for each IP network (LAN)

An important example – make sure you understand it

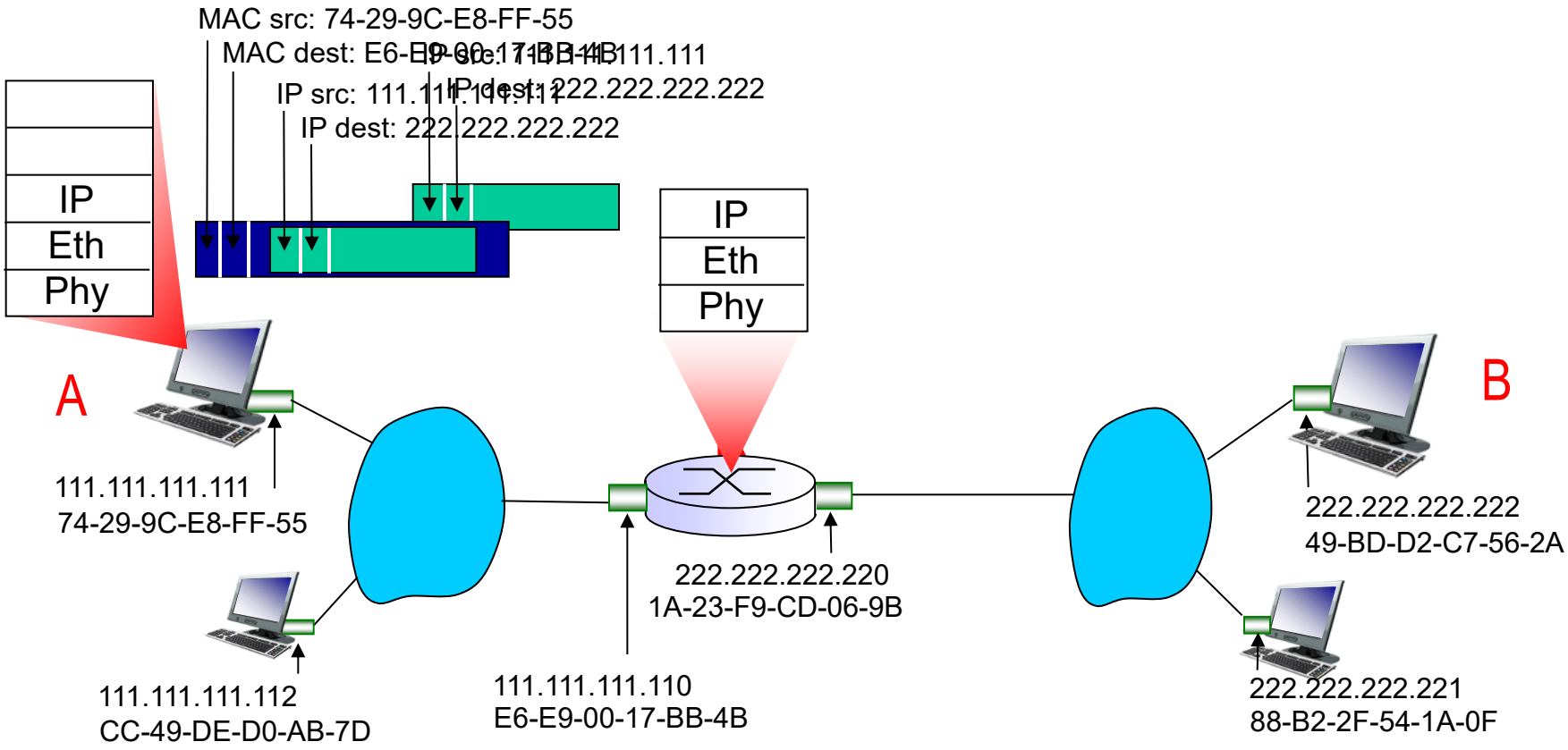
- A creates IP datagram with source A, destination B
- A uses ARP to get R's MAC address for 111.111.111.110
- A creates link-layer frame with R's MAC address as dest, frame contains A-to-B IP datagram
- A's links layer protocol sends frame
- R's links layer protocol receives frame
- R removes IP datagram from Ethernet frame, sees its destined to B
- R uses ARP to get B's MAC address
- R creates frame containing A-to-B IP datagram sends to B



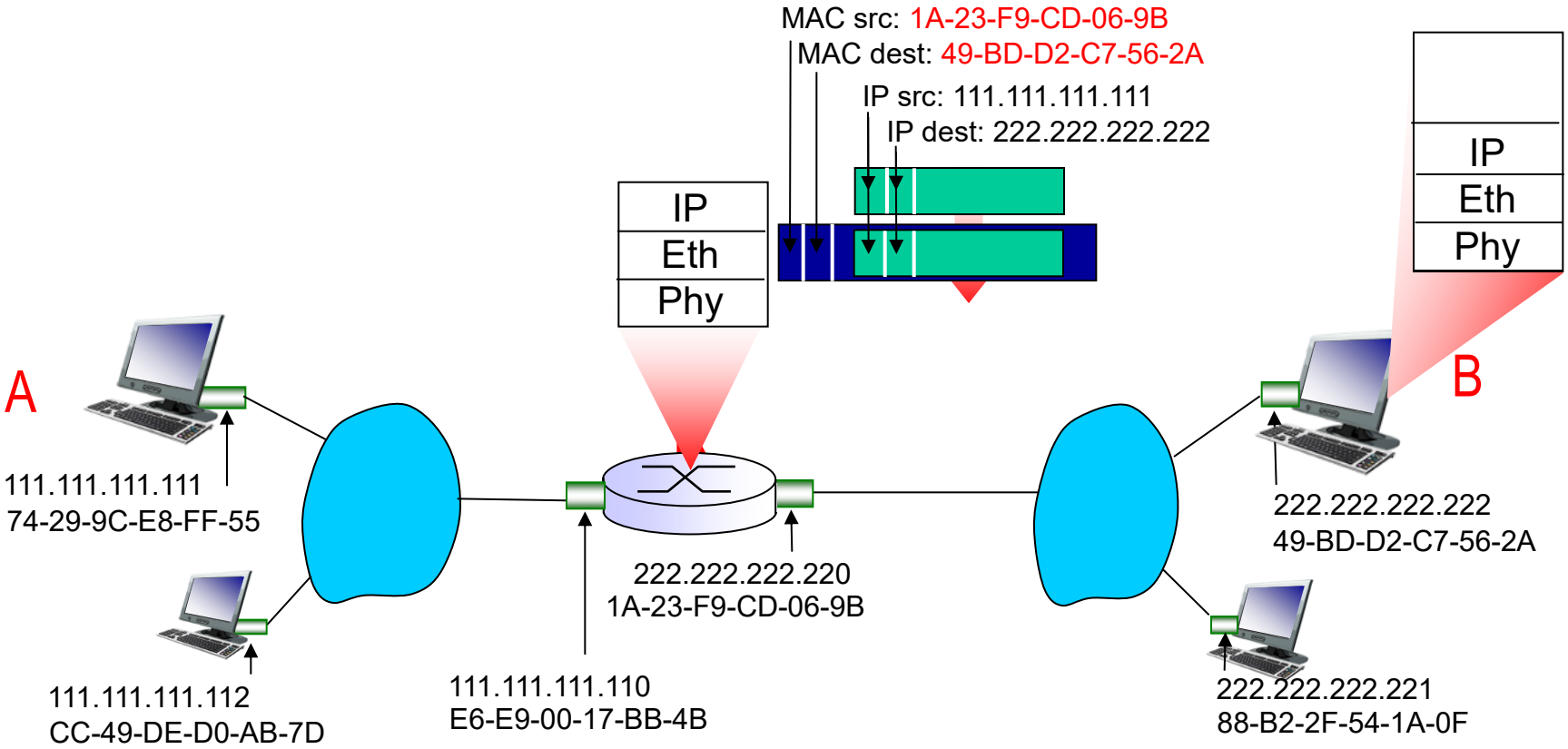
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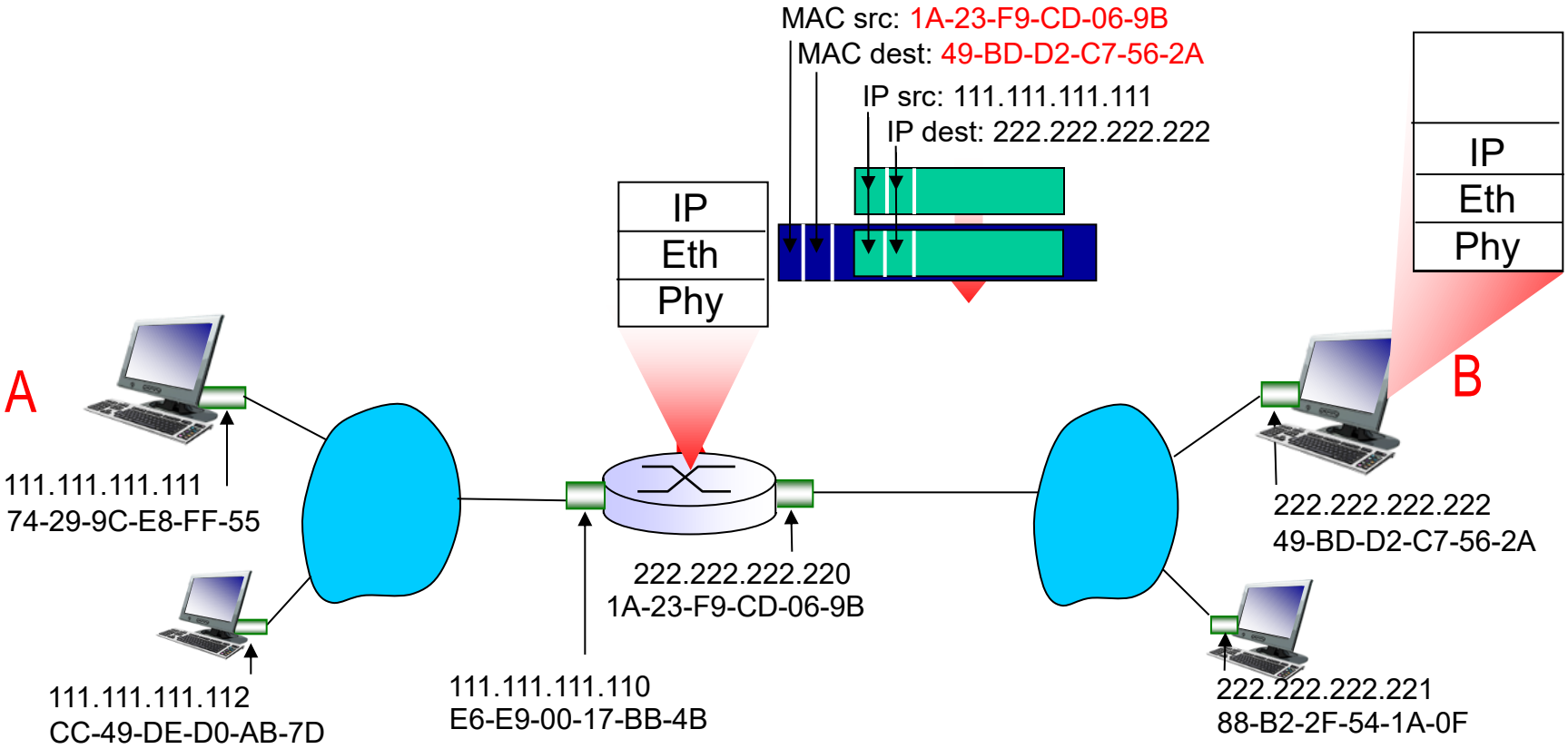
Addressing: routing to another LAN



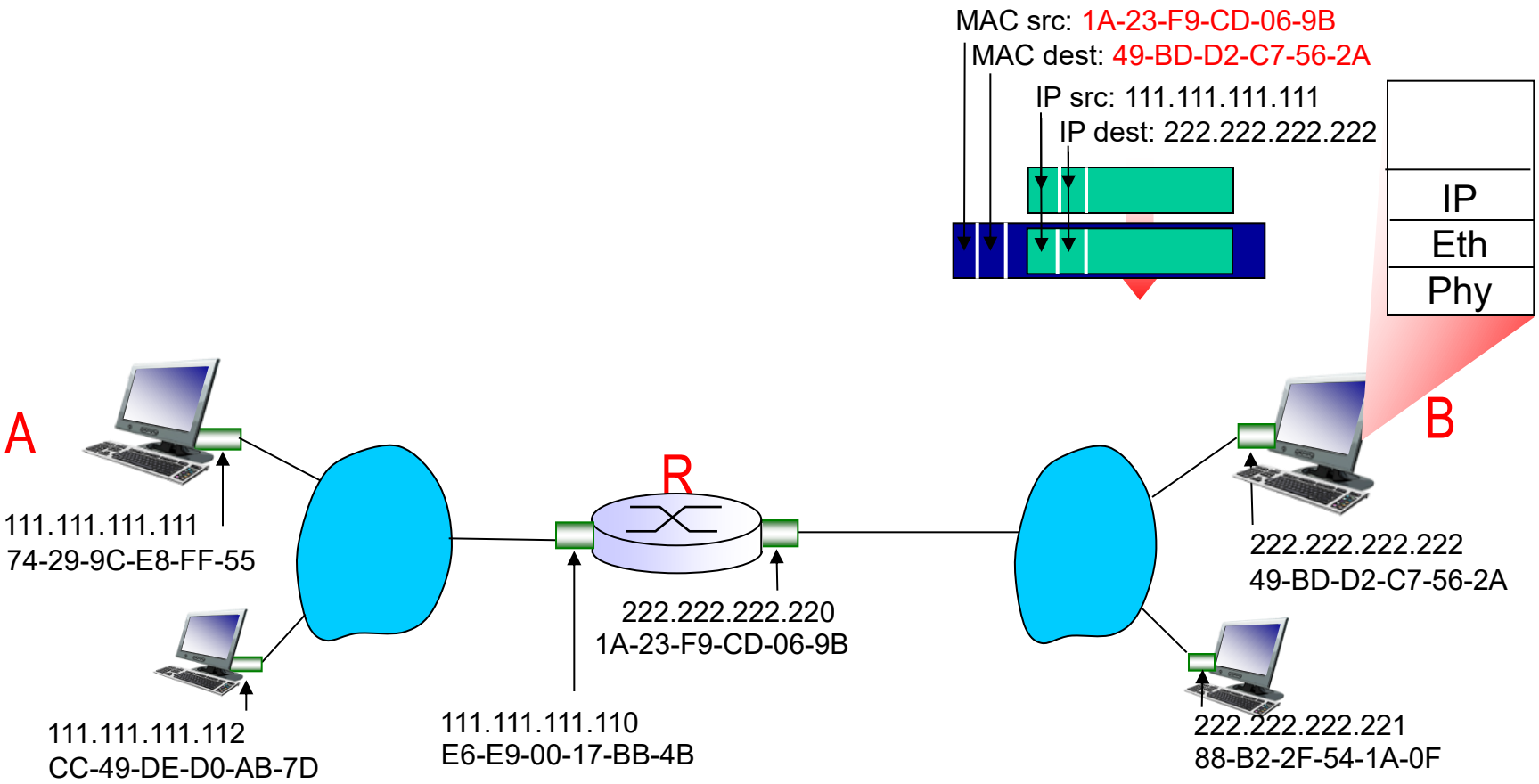
Addressing: routing to another LAN



Addressing: routing to another LAN



Addressing: routing to another LAN



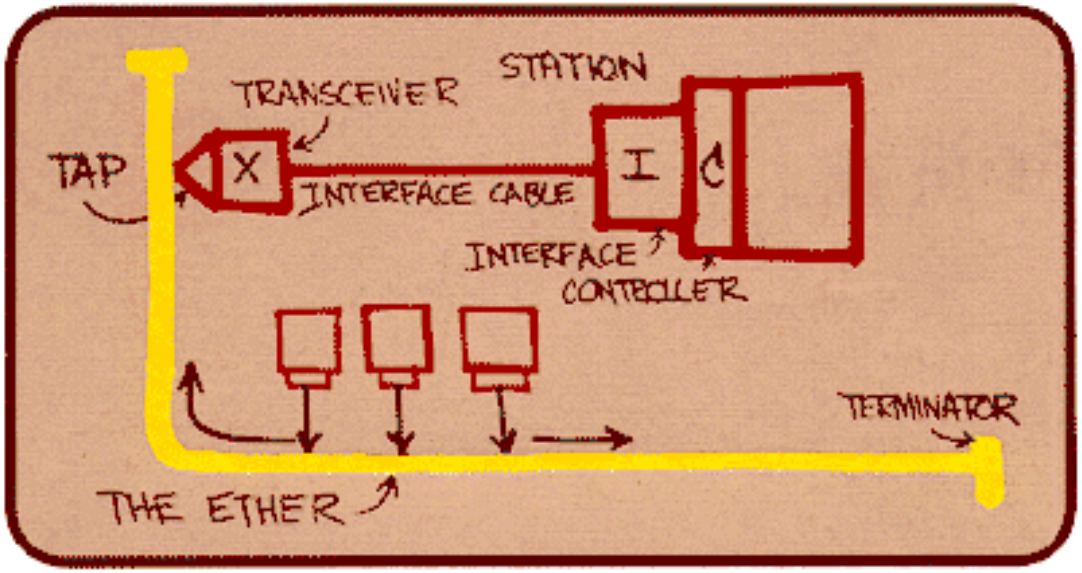
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Ethernet

"dominant" LAN technology:

- cheap $20 for NIC
- first widely used LAN technology
- Simpler, cheaper than token LANs, etc.
- Kept up with speed race: 10 Mbps – 10 Gbps



Metcalfe's Ethernet sketch

Ethernet Frame Structure

- Preamble: 7 bytes with pattern 10101010 followed by one byte with pattern 10101011 (used to synchronize receiver, sender clock rates)
- Addresses: 6 bytes
  - if adapter receives frame with matching destination address, or with broadcast address (eg ARP packet), it passes data in frame to net-layer protocol
  - otherwise, adapter discards frame
- Type: indicates the higher layer protocol, mostly IP but others may be supported (such as ATM, Novell IPX, AppleTalk)
- CRC: checked at receiver, if error is detected, the frame is simply dropped



Unreliable, connectionless service

- Connectionless: No handshaking between sending and receiving adapter.
- Unreliable: receiving adapter doesn't send acks or nacks to sending adapter
  - stream of datagrams passed to network layer can have gaps
  - gaps will be filled if app is using TCP
  - otherwise, app will see the gaps

Ethernet uses CSMA/CD

- carrier sense: adapter doesn't transmit if it senses that some other adapter is transmitting.
- collision detection: transmitting adapter aborts when it senses that another adapter is transmitting.
- random access: Before attempting a retransmission, adapter waits a random time.

Ethernet CSMA/CD algorithm

1. Adaptor gets datagram from and creates frame
2. If adapter senses channel idle, it starts to transmit frame. If it senses channel busy, waits until channel idle and then transmits
3. If adapter transmits entire frame without detecting another transmission, the adapter is done with frame !

4. If adapter detects another transmission while transmitting, aborts and sends jam signal
5. After aborting, adapter enters exponential backoff: after the mth collision, adapter chooses a **K** at random from $\{0,1,2,\ldots,2^m-1\}$. Adapter waits K*512 bit times and returns to Step 2

Ethernet's CSMA/CD (more)

Jam Signal: make sure all other transmitters are aware of collision; 48 bits;

Bit time: 1 microsec for 10 Mbps Ethernet ;
for K=1023, wait time is about 50 msec

Exponential Backoff:

- *Goal* : adapt retransmission attempts to estimated current load
  - heavy load: random wait will be longer
- first collision: choose K from {0,1}; delay is K x 512 bit transmission times
- after second collision: choose K from {0,1,2,3}…
- after 10 collisions, choose K from {0,1,2,3,4,…,1023}

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Random MAC address

MAC randomization is the process of choosing a random MAC address to connect to a network, and hiding the device's original MAC address, to maintain anonymity.

Some operating systems generate random MAC addresses when connecting to a LAN to increase privacy.

MAC randomization helps ensure the privacy of a mobile device by concealing the original MAC address, making it significantly harder to track a device based on its MAC address.

- Most phones updated to Android 10 or later have MAC randomization enabled by default.
- Apple requires users to enable MAC randomization manually.

A question from an exam

Circle the correct answer

שאלה מבחינה

הקף בעיגול את התשובה המתאימה

In Ethernet, after the fifth collisions, the probability that a node chooses K=4 is:

באיטרנט, אחרי חמש ההתנגשות ברצף, ההסתברות שהצומת המשדר יבחר K=4 היא:

1. 1/4.
2. 1/5.
3. 1/16.
4. 1/25.
5. None of the above.

1. .1/4
2. .1/5
3. .1/16
4. .1/25
5. כל התשובות (1 עד 4) לא נכונות.

A question from an exam

Circle the correct answer

שאלה מבחינה

הקף בעיגול את התשובה המתאימה

In Ethernet, after four collisions in a row between nodes A and B, the probability for an immediate fifth collisions between A and B is :

באיטרנט, אחרי ארבע ההתנגשות ברצף של שני צמתים A ו- B, ההסתברות להתנגשות מיידית חמישית בין A ו- B היא:

1. 1/4.
2. 1/5.
3. 1/16.
4. 1/25.
5. None of the above.

.1 .1/4

.2 .1/5

.3 .1/16

.4 .1/25

.5 כל התשובות (1 עד 4) לא נכונות.

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

שרת DHCP מחזיר למחשב שבקש זאת כתובת IP ובנוסף מחזיר את כתובת ה- MAC של שרת ה- DNS.

1. נכון
2. לא נכון

A DHCP server offers an IP address to the requesting host, and in addition sends the MAC address of the DNS server.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

שאילתת ARP תמיד נשלחת עם כתובת broadcast. תגובה לשאילתת ARP אף פעם לא נשלחת עם כתובת broadcast.

1. נכון
2. לא נכון

An ARP query is always sent by using a broadcast address. The ARP response is never sent with a broadcast address.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

פרוטוקול IP משמש בין היתר להעברת הודעות ARP.

1. נכון
2. לא נכון

IP protocol is used, among other things, to pass ARP messages.

1. True
2. False

Question from an exam
Circle the correct answer

שאלה מבחינה (3 נקודות)
הקף בעיגול את התשובה המתאימה

התפקיד של פרוטוקול ה- ARP הוא להשיג את כתובת ה- IP של ממשק (interface) בהינתן הכתובת הפיזית שלו .

1. נכון
2. לא נכון

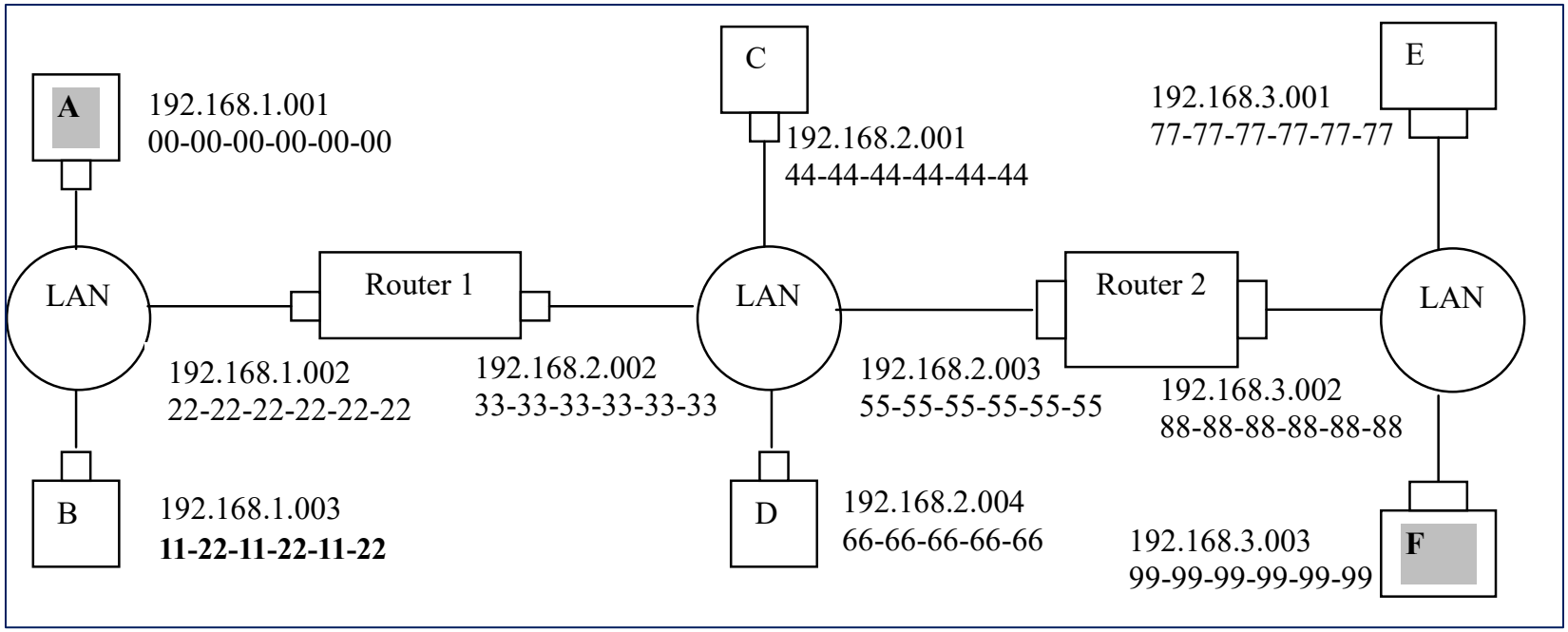
ARP is used in order to find the interface IP address based on its physical address.

1. True
2. False

Question from an exam

שאלה מבחינה (14 נקודות)

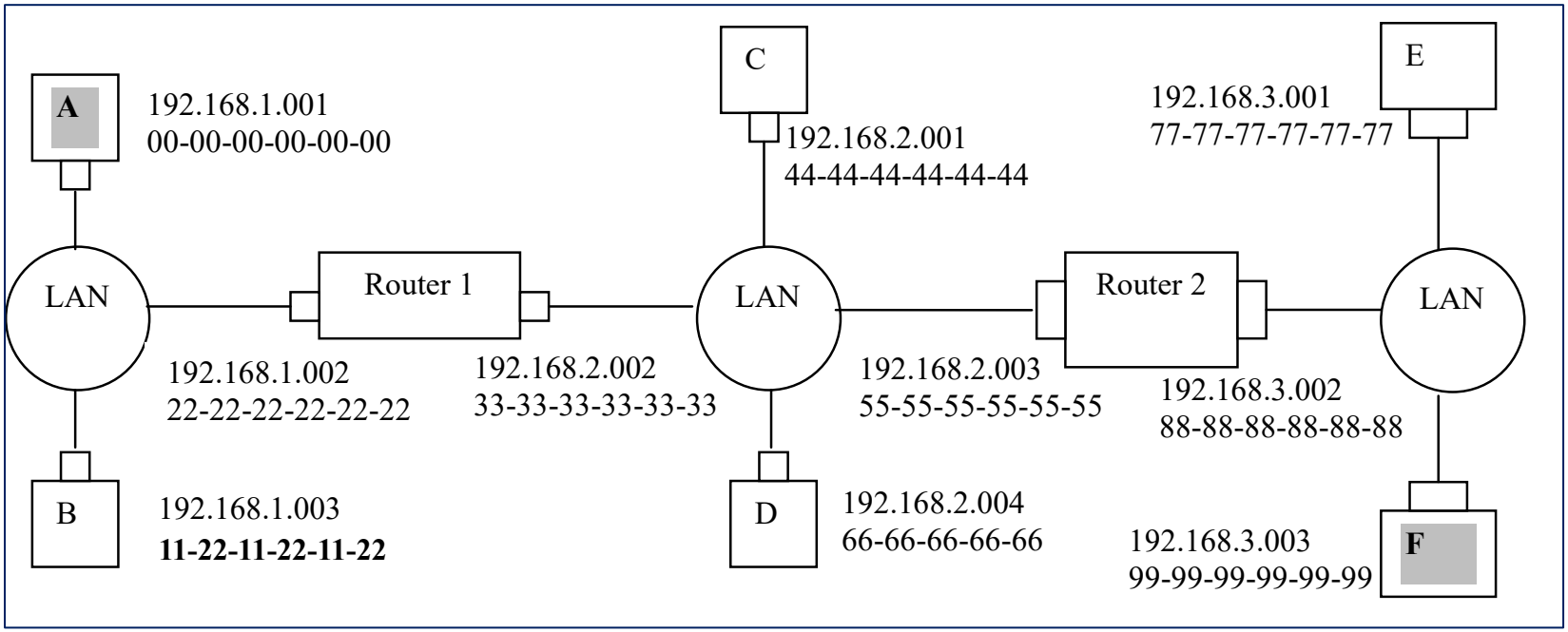
השאלה מתייחסת לרשת בתמונה בהמשך. הנח שיש לשלוח חבילה אחת ממחשב A ל- מחשב F.
הנח שכל טבלאות ה- ARP ריקות וכל טבלאות הניתוב מעודכנות. רשום בטבלה בהמשך את כל החבילות שנשלחות עד שהחבילה מגיעה ליעד. ברישום בטבלה שמור על סדר השליחה. הערה: מספר השורות בטבלה לא בהכרח משקף את המספר הכולל של חבילות שנשלחו.

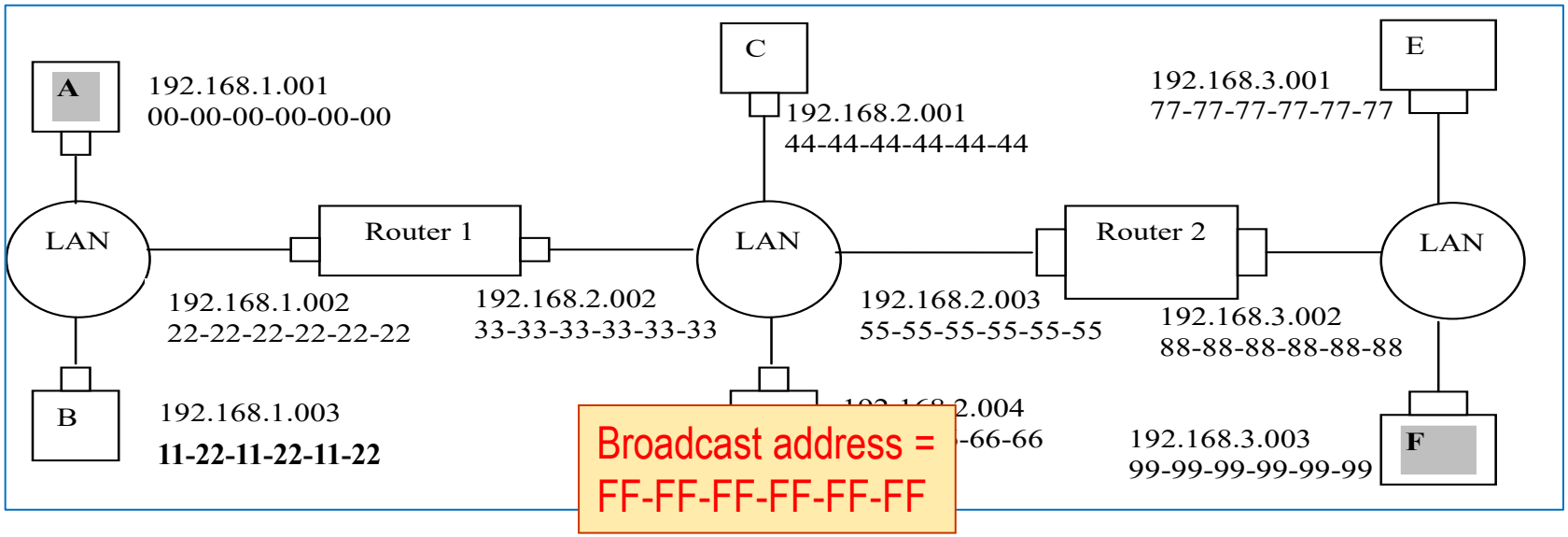


Question from an exam

שאלה מבחינה (14 נקודות)

Consider the network presented in the next figure. Assume there is one packet to be sent from host A to host F. Assume all ARP tables are empty, but all forwarding tables are updated. Fill in the (next slide) table the header details of all the packets which are sent until our desired packet finally reaches the destination. Write the records in their correct order. Note: the number of rows in the table does not necessarily represent the number of packets which are sent.





Frame		MAC Address		IP Address	
#	Type IP/ARP	source	destination	source	destination
1	ARP	A 00-00-00-00-00-00	FF-FF-FF-FF-FF-FF		
2	ARP	R1-left 22-22-22-22-22-22	A 00-00-00-00-00-00		
3	IP	A 00-00-00-00-00-00	R1-left 22-22-22-22-22-22	A	F
4	ARP	R1-right 33-33-33-33-33-33	FF-FF-FF-FF-FF-FF		
5	ARP	R2-left 55-55-55-55-55-55	R1-right 33-33-33-33-33-33		
6	IP	R1-right 33-33-33-33-33-33	R2-left 55-55-55-55-55-55	A	F
7	ARP	R2-right 88-88-88-88-88-88	FF-FF-FF-FF-FF-FF		
8	ARP	F 99-99-99-99-99-99	R2-right 88-88-88-88-88-88		
9	IP	R2-right 88-88-88-88-88-88	F 99-99-99-99-99-99	A	F

Today

- ✓ Link Layer – Introduction
- ✓ Error detection, error correction
- ✓ Multiple access links
- ✓ LAN
  - ✓ Addresses
  - ✓ Routing
  - ✓ ARP
- ✓ Routing between to a different LAN
- ✓ Ethernet
- ✓ Privacy note


CHINESE
RESTAURANT
ROCKIN'G
ALL YOU CAN EAT BUFFET
NOT MEAN ALL DAY BUFFET
YOU NO COME STAY 4 HOUR
YOU EAT - YOU GO HOME